

Tunable Rocky Planet Water Model:

Stage 2, Process Inventory and Coupling Table v0.1

Navigator's Log R&D. Started 3 October 2026. Companion to the Specification v0.1. Status: DRAFT. Everything here is a [CL] proposal pending CBH review. No equation is written in yet; each is added only after its form is confirmed in the cited source (evidence rule [CBH]).

Purpose

List every process the model computes, what it moves, what drives it, and how processes affect each other directly and indirectly, following the Process principle and the Formation principle in the spec. Processes that look alike at one scale stay separate. Reference numbers and symbols (‡ debated, ◇ single recent study) follow the spec's evidence catalog.

State variables

- Water reservoirs: surface liquid, ice, polar cold-trap ice, atmospheric vapor, groundwater and hydrothermal fluid, hydrated crust, mantle, lost to space.
- Carbon reservoirs: atmospheric CO₂, carbonate rock, mantle.
- Tracers: D/H in each water reservoir; potassium, for formation test F5.
- Thermal state: mantle temperature, core temperature, surface temperature.
- Formation outputs: planet mass and core size (tested by F1 to F4).

Process inventory

ID	Process	Moves	Driven by	Refs
Stage A: coupled formation				
P1	Accretion from the shared supply of building material	Shared supply to each planet	Giant-planet scenario, distance from the Sun	‡2, ‡3, ‡5, 14, ‡50
P2	Water delivery by bodies scattered inward	Outer material to planet water	Giant-planet growth	‡6, 7, 13
P3	Mantle stripping by collisions	Mantle rock to lost	Collision scenario	‡94, ‡95, ‡96
P4	Core formation (metal separating from rock)	Metal to core	Composition, heating	93, 97
P5	Nebular hydrogen reacting with a magma ocean	Hydrogen to water	Early hydrogen envelope	‡15, ◇61

ID	Process	Moves	Driven by	Refs
P6	Volatile retention or loss during formation	Potassium and water kept or lost	Formation heating, collisions	45, 98, ‡94
Stage B: interior				
P7	Mantle convection and cooling (stagnant or moving lid)	Heat from interior to surface	Interior heat, viscosity, lid style	68, 69, 70, 71, 36
P8	Core cooling and dynamo	Heat from core to mantle	Core-mantle heat flow	68, 31, ‡32, ‡33, ◇49
P9	Volcanic outgassing	Mantle water and CO ₂ to atmosphere	Melting, mantle temperature	36 (equation source needed)
P10	Regassing by plate recycling (moving lid only)	Surface water and carbonate to mantle	Plate motion	20 (equation source needed)
Stage C: climate and water cycle				
P11	Solar luminosity change over time	Energy input	Solar evolution	17
P12	Greenhouse warming	Sets surface temperature	CO ₂ , water vapor, hydrogen	‡39, ‡40, 38
P13	Phase partition: liquid, ice, vapor, polar cold traps	Water among surface reservoirs	Surface and polar temperature	92 (cold-trap source needed)
P14	Surface weathering in the carbonate-silicate feedback	Atmospheric CO ₂ to carbonate	Temperature, liquid water	20
P15	Underground carbonation of olivine	CO ₂ -rich groundwater to carbonate	Groundwater chemistry, not climate	◇44, 42
P16	Crustal hydration	Surface water and groundwater to hydrated crust	Water-rock contact	‡29
P17	Hydrothermal circulation	Groundwater through hot rock	Interior heat	◇44
Stage D: loss to space				
P18	Jeans (thermal) escape of H and D	Upper-atmosphere hydrogen to lost	Exobase temperature, gravity	65, 72, 73
P19	Diffusion-limited hydrogen supply	Caps P18 by how fast hydrogen rises	Water reaching high altitude	66, 28, ◇51
P20	Energy-limited early escape	Bulk atmosphere to lost	Young Sun's ultraviolet output	67, 18, 19
P21	Photochemical	Oxygen and other	Ultraviolet	Source needed

ID	Process	Moves	Driven by	Refs
	escape	atoms to lost	chemistry	
P22	Ion pickup by the solar wind	Ions to lost	Solar wind, magnetic setting	22, ‡34
P23	Sputtering	Neutral atoms to lost	Returning pickup ions	◇53
P24	HCO ⁺ dissociative recombination (Venus)	Hydrogen to lost	Ionosphere chemistry	◇56
P25	Late impacts: delivery and erosion	Water in or atmosphere out	Giant-planet instability	‡3, ‡50
P26	D/H enrichment by escape	Raises D/H of remaining water	Hydrogen escaping faster than deuterium	21, 29, ‡54 (equation source needed)

Coupling table

Sign: + means the first process strengthens the second; - means it weakens it; S means it switches the second on or off; ‡ means the sign itself is debated.

ID	From	To	Sign	Through	Refs
C1	P11 luminosity	P12 surface temperature	+	More sunlight absorbed	17
C2	P12 temperature	P14 weathering	+	Faster weathering draws down CO ₂ , which cools the surface: a damping loop (the thermostat)	20
C3	P7 lid style	P10 regassing	S	A stagnant lid shuts off recycling, which also breaks the C2 thermostat over time	36, 20
C4	P7 cooling	P8 dynamo	-	Core heat flow falls below what a dynamo needs	68
C5	P8 dynamo	P22 ion pickup	‡	Whether a magnetic field protects the atmosphere is	‡34

ID	From	To	Sign	Through	Refs
				debated	
C6	P7 interior heat	P9 outgassing	+	A cooling mantle melts less and outgasses less	36
C7	P12 temperature	P19 hydrogen supply	+	A warmer atmosphere carries more water high up, raising escape; this loop amplifies	28, 51
C8	P11 young Sun ultraviolet	P20 early escape	+	More ultraviolet energy drives bulk escape	18, 19, 67
C9	P18 to P24 escape	P26 D/H	+	Escape removes hydrogen faster than deuterium	21, 29
C10	P16 crustal hydration	P26 D/H	none	Water locked in rock leaves D/H unchanged; this is what lets D/H tests separate escape from storage	29
C11	P1 giant-planet scenario	P1 on other planets	-	Material taken by one planet is unavailable to the others; Jupiter can starve the Mars region	2, 5, Formation principle
C12	P1 planet mass	P18 to P24 escape	-	Higher mass means higher escape velocity and less escape	65
C13	P1 planet mass	P7 cooling time	+	Larger planets stay hot longer, which keeps P8 and P9 running longer	68
C14	P3 mantle	P4 core fraction	+	Losing mantle	94, 95, 45

ID	From	To	Sign	Through	Refs
	stripping			rock raises the core's share; possible volatile loss through P6 is debated	
C15	P13 polar cold traps	Mercury surface water	+	Shadowed craters keep ice despite a hot surface	92
C16	P15 olivine carbonation	P12 temperature	-	Draws down CO2 without a temperature feedback, so it cools without acting as a thermostat	◇44
C17	P7 interior heat	P17 hydrothermal circulation	+	Hot rock drives groundwater circulation	◇44
C18	P22 ion pickup	P23 sputtering	+	Returning pickup ions knock neutral atoms loose	◇53
C19	P5 nebular hydrogen	Mantle water	+ ‡	Water made from hydrogen and magma; contested	‡15, ◇61

Gaps before equations

- Equation sources still needed for P9 outgassing, P10 regassing, P13 cold traps, P21 photochemical escape, and the form of P26 D/H enrichment.
- Every equation for P1 to P26 will be written only after its form is confirmed in the cited source; unconfirmed forms are not used.
- Independence safeguard (spec): P1 imports mechanisms from published simulations, never their final masses or core sizes.

Verified equations, block 1 [CL, pending CBH review]

Method: each equation's form is confirmed in at least one retrievable source, then checked against numbers that source itself reports. Form sources (FS) are listed at the end of this section; they confirm how an equation is written, not new evidence values.

- E1, solar luminosity (P11): $L(t) = L_{\text{now}} / [1 + (2/5)(1 - t/t_{\text{sun}})]$, with $t_{\text{sun}} = 4.57$ Gyr. Source: ref 17, form confirmed in FS1. Check: 0.714 of today at the Sun's birth; 0.765 at 3.5 billion years ago (source: about 76 percent); 0.942 at 0.7 billion years ago (source:

about 94 percent). Pass. Caveat: FS1 reports good agreement with standard solar models except the first ~0.2 Gyr.

- E2, Jeans escape (P18): $\text{flux} = n u / (2\sqrt{\pi}) \times (1 + \lambda) e^{(-\lambda)}$, with $u = \sqrt{(2kT/m)}$ and $\lambda = GMm / (kT r_{\text{exo}})$ at the exobase. Source: ref 65, form confirmed in FS2. Check: for Earth's exobase at 1000 K, λ is about 7 for hydrogen, 28 for helium and 112 for oxygen, so escape speed divided by thermal speed is 2.7, 5.3 and 10.6; FS10 states Jeans escape matters on Earth only when that ratio is below about 6, true only for hydrogen and helium. Pass. Caveat: a lower limit; the exobase height and temperature dominate the result.
- E3, diffusion-limited hydrogen supply (P19): $\text{flux} = \Sigma N_i f_i b_i / H_a$ (light gas through a heavier background; general form $b f (1/H_a - 1/H_i)$). Source: ref 66, form confirmed in FS3 and FS4. Check: with the published H₂ to N₂ diffusion parameter of 1.7×10^{21} per meter per second, the reported Earth flux of 1.5×10^{12} per square meter per second requires about 5 ppmv of hydrogen-bearing gas, the same order as measured stratospheric water (2.5 to 4.5 ppmv). Pass at order of magnitude. Caveat: an upper bound on hydrogen loss; a newer Mars-specific formulation is reported to be more accurate (FS11).
- E4, energy-limited early escape (P20): $dM/dt = \epsilon \pi F_{\text{XUV}} R_{\text{XUV}}^2 R_p / (G M_p K_{\text{tide}})$. Source: ref 67, form confirmed in FS5 and FS6. Check: pending; planned test is to reproduce a published Venus water-loss time. Caveat: the heating efficiency ϵ is not known in advance, so the result is an upper limit or order-of-magnitude estimate; ϵ stays a tunable input with its range.
- E5, mantle convection (P7): moving lid $q = (k/d)(T_p - T_s)(Ra/Ra_c)^\beta$; stagnant lid $q = (k/2d)(T_p - T_s) \theta^{(-4/3)} Ra^{(1/3)}$, with $\theta = E (T_p - T_s) / (R T_p^2)$ and $Ra = \alpha g \Delta T d^3 / (\kappa \nu)$. Convective speed under a stagnant lid: $u = 0.38 (\kappa/h) (Ra/\theta)^{(1/2)}$, which drives the lava-lamp motion. Sources: refs 69, 70; forms confirmed in FS7 and FS8. Check: pending until the viscosity range is set in stage 3. Caveat $\pm$: melting and crust can lower heat flux by up to a factor of about 5 to 10 (FS9), and mantle viscosity is uncertain by orders of magnitude.

Form sources: FS1 Feulner (2012) The faint young Sun problem, Reviews of Geophysics, arXiv:1204.4449. FS2 Atmospheric escape processes and planetary atmospheric evolution (2020), arXiv:2003.03231. FS3 The cosmic shoreline (2017), arXiv:1702.03386. FS4 Effect of mantle oxidation state and escape upon the evolution of Earth's magma ocean atmosphere, arXiv:2009.14599. FS5 Energy-limited escape revised, arXiv:1511.09348. FS6 A critical assessment of the applicability of the energy-limited approximation, arXiv:2105.05858. FS7 Was Venus ever habitable? Constraints from a coupled interior-atmosphere-redox evolution model, arXiv:2111.00033. FS8 Terrestrial planet evolution in the stagnant-lid regime, arXiv:1210.3838. FS9 Korenaga (2009) Scaling of stagnant-lid convection with Arrhenius rheology. FS10 Survival of primordial planetary atmospheres: mass loss from temperate terrestrial planets, arXiv:1912.08820. FS11 Diffusion limited escape of hydrogen from Mars (Yelle, University of Arizona LPL preprint). Authors are listed only where the retrieved page showed them.

Verified equations, block 2 [CL, pending CBH review]

- E6, thermal dynamo criterion (P8): a thermally driven dynamo needs core-mantle boundary heat flow above the heat carried by conduction along the core's adiabat, $Q_{\text{ad}} = 4\pi r_{\text{cmb}}^2 k \alpha g T_{\text{cmb}} / c_p$. Source: ref 68; form confirmed in FS12 and FS13.

Check: Q_{ad} scales linearly with core conductivity k ; FS14's 8 TW at $k = 50$ W/m/K becomes 21 to 32 TW at 130 to 200 W/m/K, consistent with FS15's statement that 2 to 4 times higher conductivity gives more than 15 TW. Pass (consistency). Caveat $\pm$: core thermal conductivity is actively debated, so k is a tunable input with that full range.

Process note: chemical convection from a growing inner core can drive a dynamo even below Q_{ad} (FS13), a look-alike process kept separate under the Process principle.

- E7, D/H enrichment by escape (P26): Rayleigh distillation. With hydrogen lost as $dH/dt = -kH$ and deuterium as $dD/dt = -f k D$, the ratio grows as $R/R_0 = (H/H_0)^{(f-1)}$, so initial water = present water $\times (R/R_0)^{1/(1-f)}$. Sources: refs 21, 29; form confirmed in FS2 and FS16. Check: for Venus at about 100 times Earth's D/H, the initial-to-present water ratio is 100 with $f = 0$, about 200 with $f = 0.13$, and 100,000 with $f = 0.6$, so f controls the answer more than the D/H measurement does. FS16 reports f between 10^{-1} and 10^{-5} for most Mars conditions, about 0.06 when non-thermal escape is included. Caveat: f changes with escape process and era, and the starting D/H is itself uncertain; f is tracked through time rather than fixed.
- E8, photochemical oxygen escape (P21): escape rate scales with solar EUV as a power law, rate $\propto (EUV)^n$. Two verified values for n disagree $\pm$: $n = 2.6 \pm 0.6$ from MAVEN in situ data (FS17) and $n = 1$ from an analytical treatment (FS18); other studies did not confirm the steep exponent at higher EUV (FS19). The model carries n from 1 to 2.6 as a tunable range. Check: FS17's modern average of 4.3×10^{25} oxygen atoms per second equals 1.14 kg/s, which is 38 to 57 percent of the 2 to 3 kg/s total Mars escape in ref 27, consistent with FS17 calling it a dominant channel. Pass. Planned check: reproduce FS17's 13, 49, 189 and 483 mbar oxygen loss over 1, 2, 3 and 3.5 Gyr.
- E9, polar cold-trap stability (P13): water ice is retained where temperatures stay low enough that ice sublimates to vacuum at under 1 mm per billion years; MESSENGER altimeter brightness matches the thermal model's predicted ice locations (FS20). Source: ref 92 with FS20. In a global box model this is a stability rule plus a cold-trap area input, not a spatial map. Check: qualitative match only (Me2 requires presence, not amount).
- E10, surface-mantle water exchange with moving plates (P9 and P10 for Earth): degassing at ridges = spreading rate $\times$ mantle water content $\times$ melt depth $\times$ fraction degassed, where the fraction falls with seafloor pressure; regassing by subduction = spreading rate $\times$ water in the hydrated layer $\times$ layer thickness $\times$ return efficiency. Source: ref 20 for the setting; forms confirmed in FS21 and FS22 (after Cowan and Abbot 2014). Caveat $\pm$: whether degassing and regassing depend mainly on pressure or on mantle temperature changes the long-term ocean size (FS21), so both versions are kept as a selectable option. Stagnant-lid outgassing for Venus, Mars and Mercury still needs its own source (block 3).

Form sources, block 2: FS12 Fluid dynamics of Earth's core: geodynamo, inner core dynamics, core formation, arXiv:1905.10416. FS13 Thermal conductivity and adiabatic heat flux study, Scientific Reports, doi:10.1038/s41598-022-14130-z. FS14 Core thermal conductivity study reporting 8.0 TW adiabatic CMB heat flow, NSF public access record 10215002. FS15 Gubbins, Alfe, Davies, Pozzo (2015) On core convection and the geodynamo: effects of high electrical and thermal conductivity, UCL Discovery 1498724. FS16 Higher Martian atmospheric temperatures at all altitudes increase the D/H fractionation factor and water loss, arXiv:2007.12589. FS17 Lillis, R.J. et al. (2017) Photochemical escape of oxygen from Mars: first results from MAVEN in situ data, JGR Space Physics 122(3):3815-3836. FS18 Cravens, T.E. et al. (2017) Hot oxygen escape from Mars: simple scaling with solar EUV irradiance, JGR. FS19 Upper atmosphere dynamics and drivers of volatiles loss from terrestrial-type

(exo)planets, arXiv:2602.15507. FS20 Paige, D.A. et al. (2013) Thermal stability of volatiles in the north polar region of Mercury, Science 339:300, doi:10.1126/science.1231106. FS21 Effect of surface-mantle water exchange parameterizations on exoplanet ocean depths, arXiv:1609.04786. FS22 The persistence of oceans on Earth-like planets: insights from the deep-water cycle, arXiv:1501.00735.

Verified equations, block 3 [CL, pending CBH review]

- E11, silicate weathering (P14): $W/W_0 = (P/P_0)^{0.3} \times \exp((T - 285 \text{ K}) / 13.7 \text{ K})$, where P is the CO₂ partial pressure and the 0 subscripts are present-day values. Source: ref 20, read in the original paper this block. The 13.7 K combines a 17.7 K rock-dissolution term with a weak 60 K runoff term. Correction found: a 2026 paper restating this formula uses 288 K as the reference temperature; the original uses 285 K, and the model uses the original. Caveats from the authors themselves: the quantitative details are speculative, and the CO₂ dependence is extrapolated from experiments at 2 to 20 bar. Alternatives (a runoff-limited model and a supply-limited ceiling) are kept as selectable options ‡. The original also notes weathering effectively stops below freezing.
- E12, greenhouse warming near present Earth (P12): $T_s = T_e + T_g(\text{H}_2\text{O}) + T_g(\text{CO}_2)$, with $T_g(\text{H}_2\text{O}) = 29.7 + 0.5 (T_s - 285)$ and $T_g(\text{CO}_2) = 2.3 (P/P_0)^{0.364}$, combined as $\Delta T = 2 \Delta T_e + 4.6 (P/P_0)^{0.364} - 4.6$. Source: ref 20 (original). Checks: a 25 percent fainter Sun lowers T_e by 17.6 K (paper: 18 K); solving the coupled weathering and greenhouse equations gives $\Delta T = -18.0 \text{ K}$, matching the paper, versus -36 K with no feedback (paper: the feedback cuts the change by about half); a CO₂ doubling gives 1.32 K (paper: 1.3 K). Pass. Note: the paper's stated 265 K does not equal its own $285 - 18 = 267 \text{ K}$, a small internal inconsistency, recorded and not corrected. Limit: valid only near present Earth conditions; the authors note it underestimates CO₂ warming compared with detailed models (2 K per doubling). Thick-CO₂ atmospheres (Venus, early Mars) need a different scheme (block 4).
- E13, runaway greenhouse threshold (P12, P13): surface oceans cannot persist when absorbed sunlight exceeds the thermal radiation limit of 282 W/m² (FS24). Checks: Earth absorbs 240 W/m² today (albedo 0.294, ref 91), below the limit, consistent with its oceans. Venus with an Earth-like albedo of 0.3 absorbs 456 W/m² today and 325 W/m² under the young Sun, a runaway in both cases. So early Venus oceans are possible only if clouds gave it a much higher albedo, which is exactly where ‡23 and ‡24 disagree. Mars stays far below the limit. Caveat: FS24 states that avoiding a runaway requires an atmosphere below saturation and clouds that cool more than they warm, so cloud treatment can move this threshold.
- E14, outgassing on single-shell planets (P9 for Venus, Mars and Mercury): model structure confirmed in FS25 (thermal evolution, crust production and CO₂ cycling together), but the equation forms were not retrieved this block, so E14 is pending and not used. Verified findings from FS25 that shape couplings: when weathering is limited by the supply of fresh rock, outgassing can overwhelm drawdown and build a hot CO₂-rich atmosphere, and the carbon a planet accretes during formation is critical to whether it stays temperate.
- E8 reproduction check, now run: with $n = 2.6$ and the young Sun's EUV falling as $t^{-1.23}$, the model gives 13.9, 49.0, 182 and 449 mbar of oxygen lost over 1, 2, 3 and 3.5 Gyr, against FS17's 13, 49, 189 and 483 (within about 7 percent). Pass. With $n = 1$ the 3.5

Gyr loss is 73 mbar, about 6.6 times less, so the $\pm$ choice of n matters. Caveat: the $\tau^{-1.23}$ exponent is ref 18's power law as quoted in FS26; it will be confirmed in ref 18 directly in block 4.

Proposed new couplings [CL]: C20, crust production (P9) caps weathering (P14) on single-shell planets (supply limit), FS25. C21, carbon accreted at formation (P1, P2) sets the long-term climate of single-shell planets, FS25; this links your Formation principle directly to climate. C22, absorbed sunlight (P11 with albedo) switches surface oceans off above 282 W/m² (sign S), FS24.

Form sources, block 3: ref 20 (Walker, Hays and Kasting 1981) read in full from the original. FS24 Goldblatt, C., Robinson, T.D., Zahnle, K.J., Crisp, D. (2013) Low simulated radiation limit for runaway greenhouse climates, Nature Geoscience 6:661-667, doi:10.1038/ngeo1892. FS25 Foley, B.J., Smye, A.J. (2018) Carbon cycling and habitability of Earth-size stagnant lid planets, Astrobiology 18(7):873-896, arXiv:1712.03614. FS26 On the role of dissolved gases in the atmosphere retention of low-mass low-density planets, arXiv:1802.04296.

Verified equations, block 4 [CL, pending CBH review]

- E16, young Sun's high-energy output (P11, P20, P21): $F_{\text{XUV}} = 29.7 \times \tau^{-1.23}$ erg/s/cm² at 1 AU for 0.1 to 120 nm, with τ in Gyr, and flat (saturated) before about 0.1 Gyr (FS27, after ref 18). Check against ref 18's own abstract: the power law gives 2.65 times today's flux 2.5 Gyr ago (ref 18: about 2.5) and 5.96 times 3.5 Gyr ago (ref 18: about 6). Pass. This closes the caveat on the E8 reproduction check. Caveat $\pm$: the curve depends on how fast the young Sun spun (ref 19), so slow, medium and fast tracks stay as a tunable input.
- E14 upgraded, outgassing on single-shell planets (P9): now partly verified. Structure confirmed in FS25's own text: melt production = melt fraction $\times$ the flow of mantle into the melt zone, with that flow through a cylindrical upwelling equal to $v (d_m - \delta) \pi L$, scaled to the whole planet; outgassing = melt production $\times$ volatile content $\times$ degassing efficiency, plus CO₂ released when buried carbonate-bearing crust heats up (FS28). Weathering on these planets is capped by fresh crust: F_w is proportional to crust production $\times (P_{\text{CO}_2} / P_{\text{CO}_2, \text{Earth}})^\alpha$ (FS29), which gives coupling C20 its equation. Look-alike conflict $\pm$: two published scalings for convective speed under a lid disagree, $u = 0.38 (\kappa/h)(Ra/\theta)^{(1/2)}$ (E5) and $u = 0.05 \kappa (Ra/\theta)^{(2/3)} / d$ (FS7 after FS25); both are kept, and the choice also changes the lava-lamp motion speed. Still needed: the melt-fraction formula and constants, to be read from FS25's full text in stage 3. Check target: a published stagnant-lid model with 520 ppm CO₂ in its melt could not produce Venus's roughly 87 bar CO₂ atmosphere (FS30).
- E15, greenhouse for thick CO₂ atmospheres (P12, early Mars): form confirmed as polynomial fits to one-dimensional radiative-convective models for outgoing infrared and reflected sunlight, valid from 190 to 360 K and 0.00001 to 10 bar CO₂ (FS31, after Williams and Kasting 1997; FS32 with hydrogen added). Coefficients not yet retrieved, so E15 is pending. Verified constraints it must satisfy: a 3D climate model found pure CO₂ at 0.1 to 7 bar could not lift early Mars's annual mean above freezing, and CO₂ collapses into polar ice above about 3 bar (FS33); ref 39 found 1.3 to 4 bar of CO₂ with 5 to 20 percent H₂ was needed. Consequence: the model cannot pass test M6 on pure CO₂ alone; it must include hydrogen warming or episodic warming (± 39 , ± 40) or report an honest M6 failure. Venus at 92 bar is outside this fit's range, so Venus is handled by

E13 and present-state tests only.

- P1, accretion: a key design conflict, raised as decision D10. FS34 shows that putting 2 Earth masses of material in a narrow ring from 0.7 to 1.0 AU produces Venus and Earth analogs and often Mercury and Mars analogs. Critiques: Mercury rarely forms correctly in that setup (FS35: 9 percent of good runs reach the real Mercury-Venus spacing), and FS35 notes that a Mars built from the same ring should share Earth's isotopic makeup; a source comparing Mars and Earth isotopes is not yet in the catalog. The conflict: the independence safeguard forbids importing published final masses, but a box model cannot compute accretion outcomes on its own.
- P4, core formation: proposed as a mass balance, where core fraction comes from the metal content of the accreted building blocks plus any mantle lost to collisions (P3). Source for building-block iron content not yet retrieved; pending.
- Checks rescheduled: E4 (energy-limited escape against a published Venus water-loss time) moves to stage 4, where the full escape module runs; E5 (convection) moves to stage 3, where the viscosity range is set.

Decision D10 for CBH, options: A, run our own small N-body accretion simulations from mechanism inputs only (open-source code on free compute); F1 and F2 then stay fully independent tests, at real compute and setup cost. B, use a simple analytic growth law from mechanism inputs; cheap but crude, and F1 and F2 lose much of their diagnostic power. C, keep published outcomes as inputs and relabel F1 and F2 as consistency checks rather than tests, which is honest but weakens the formation tests you approved. [CL recommendation: A, because it is the only option that keeps the formation tests real.]

D10 decided [CBH, 3 October 2026]: option A. The model runs its own small N-body accretion simulations from mechanism inputs only (where the building material starts, how much there is, and the giant-planet configuration chosen in the D7 selector). D7 still governs giant-planet history; D10 governs how the rocky planets grow under it.

Feasibility smoke test [CL]: the open-source N-body code REBOUND (version 5.2.1) installed and ran in this environment. A test with the Sun, a Jupiter-mass planet and 50 embryos holding 2 Earth masses in the 0.7 to 1.0 AU ring ran 10,000 years with a relative energy error of 1.3×10^{-5} , which is good. It took 215 seconds, so at this speed 10 million years would take about 2.5 days and 100 million years about 25 days per run here. Real runs therefore need a reduced setup, a larger time step tested for accuracy, and long batches on free Colab or Kaggle compute.

Open design choices for the accretion runs [CL]: number of bodies, run length, and number of repeat runs (outcomes vary run to run, so results are reported as a spread, not a single answer). Most important: the collision model. Simple merging cannot strip a mantle, so it can never produce Mercury's oversized core; testing F3 through formation needs a collision-outcome model that includes hit-and-run and fragmentation, and that model needs its own verified source before use.

Form sources, block 4: FS27 The Sun and stars as the primary energy input in planetary atmospheres, arXiv:0911.4872. FS28 Mantle degassing lifetimes through galactic time and the maximum age stagnant-lid rocky exoplanets can support temperate climates, arXiv:2204.04243. FS29 Early habitability and crustal decarbonation of a stagnant-lid Venus, arXiv:2109.08756. FS30 Mantle degassing and the origin of the Venusian atmosphere, EPSC 2009 abstract 616. FS31 Energy balance model for early Mars, Astronomy and Astrophysics 2012, doi:10.1051/0004-6361/201118527. FS32 Climate cycling on early Mars caused by the carbonate-silicate cycle, arXiv:1609.00602. FS33 3D modelling of the early Martian climate under a denser CO₂ atmosphere, arXiv:1210.4216. FS34 Hansen, B.M.S. (2009) Formation of

the terrestrial planets from a narrow annulus, arXiv:0908.0743. FS35 Terrestrial planet formation from two source reservoirs, arXiv:2507.14814.

Accretion run design, part 1: collision model [CL, pending CBH review]

Search result, 4 October 2026: a published, open-source collision-outcome model exists that handles hit-and-run strikes and fragmentation, and it is written for REBOUND, the code chosen under D10. Its sources are refs 99 to 105 in the Specification's evidence catalog addendum.

- Physics source: Leinhardt and Stewart (2012) [99] derived scaling laws, from hydrocode and N-body impact simulations, that sort any collision between gravity-dominated bodies into cratering, merging, disruption, super-catastrophic disruption or hit-and-run, and give the masses and speeds of the bodies that come out. Applied to standard planet-formation runs, the companion paper [100] found giant impacts between embryos split roughly evenly among partial accretion, graze-and-merge, and hit-and-run, so perfect merging misrepresents most of them.
- Implementation: Childs and Steffen (2022) [102] built a fragmentation module for REBOUND from [99], following the method Chambers (2013) [101] used for the MERCURY code, and compared the two codes. It resolves eight outcomes: elastic bounce, perfect merger, effective merger, graze-and-merge, partial accretion, hit-and-run, partial erosion, and super-catastrophic disruption. The code is public (github.com/annacrn/rebound_fragmentation, GPL-3 license), was last updated in August 2026, and states compatibility with REBOUND 4.4.10 only. It writes a collision report that later steps use to track composition.
- Core and mantle: inside the N-body run every body is one uniform layer. Core and mantle are followed afterwards by a post-processing tracker [103] that reads the collision report and assigns stripped material to the mantle first. The collision code itself never sees a body's iron fraction change. The same limit will apply to the D/H and potassium tracers if they are followed this way.
- Integrator: REBOUND's TRACE integrator gained support for fragmentation, with bodies added or removed in the middle of a time step, in version 4.4.10 [105]. Its authors report speedups of up to about 70 times over MERCURIUS on a 30-body test.

Smoke test 2 [CL], 4 October 2026: REBOUND 4.4.10 (C version) compiled cleanly with the fragmentation module. The module authors' example setup (radii inflated 5 times to speed up collisions, 6-day time step, Jupiter and Saturn included) ran with collisions resolved and logged, including mergers and effective mergers. Timing in this environment: with 286 bodies, about 104 seconds per 1,000 simulated years under both MERCURIUS and TRACE. With 154 bodies, the count used in a published Mercury study [104], about 44 seconds per 1,000 years under MERCURIUS and 33 under TRACE. At the TRACE rate, 1 million years takes about 9 hours, and the published run length of 300 million years about 110 days per run on one core. That is an upper estimate, since bodies merge and the count drops, but it is far beyond free compute for the dozens of repeat runs published studies use (25 to 43 per setup in [104]). Version note: the stage 1 smoke test used REBOUND 5.2.1; formation runs should pin 4.4.10 to match the module, and the freeze hash should record it [CL].

Finding that bears on test F3 [CL]: a 2025 study ¹⁰⁴ used this same toolchain to test Mercury's formation. Starting from building blocks with the same iron fraction everywhere, erosive collisions did make iron-rich fragments early, but the stripped mantle stayed in the disk and

merged back, and the planets drifted back to their starting iron fraction. No Mercury formed. A Mercury-like core fraction appeared only when the inner disk started iron-rich. Their simulated Mercuries were also too massive, a problem they describe as common. The authors name two limits: nothing removes stripped mantle debris, and collisions too weak to make a fragment above the minimum fragment mass are treated as mergers. If this holds, our model with uniform building blocks will most likely fail F3. Under the Process principle that failure is a result to report, not a reason to adjust inputs.

Decision D11 for CBH: how building-block iron is set for the formation runs. A, the same iron fraction everywhere, taken from measured solar composition; F3 then tests whether collisions alone can make Mercury, and the expected result is a reported miss. B, iron fraction varies with distance from the Sun, set only from a published mechanism (for example iron and rock separating before planetesimals form) and never chosen to make Mercury pass; this needs its own verified source that gives a number. C, run both as named scenarios in the selector and report F3 for each, so the model shows which explanation it supports. [CL recommendation: C, because it turns Mercury's core into a comparison between two hypotheses instead of a single pass or fail.]

D11 decided [CBH, 4 October 2026]: option C. Building-block iron runs as two named scenarios in the selector. Scenario A: the same iron fraction everywhere, from measured solar composition. Scenario B: iron fraction varies with distance from the Sun, set only from a published mechanism, never chosen to make Mercury pass. F3 is reported for each scenario. Scenario B cannot run until a verified mechanism source that gives a number is in the catalog; until then only scenario A is runnable.

Leads not yet verified, not used [CL]: a single-giant-impact probability under 1 percent (Franco et al. 2022, as cited in 104); the Marcus et al. (2010) mantle-stripping rules used by 104; a hit-and-run remnant model that treats core and mantle as layers (Gabriel et al. 2020); removal of stripped debris by radiation forces (Spalding and Adams 2020, as cited in 104); and SHARD, a June 2026 preprint that interpolates collision outcomes from a catalog of hydrodynamic impact simulations and tracks water through each collision. SHARD may suit the water tracers but was read only as a summary.

Still open for part 2: run length and repeat count under the compute limit above; body count and minimum fragment mass; whether to keep the 5 times radius inflation (the module authors found no significant change up to a factor of 3, as reported in 104); and where runs are hosted. The module supports restarting from saved archives, which short free-compute sessions would need.

Accretion run design, part 2: compute and run plan

[CL, pending CBH review]

Hosting [CL proposal]: GitHub Actions on a public repository under github.com/NavigatorsLog. Usage is free for standard hosted runners in public repositories, each job can run up to 6 hours, and a free account can run 20 jobs at once (GitHub documentation, checked 4 October 2026). That fits the zero-cash mandate. Each run is split into 6-hour chunks: a chunk saves REBOUND's simulation archive and the next chunk restarts from it, which the fragmentation module supports. A public repository also lets anyone inspect the runs, which suits the falsification-first method. The runner hardware has not been timed yet, and OpenMP parallel gravity could not be tested here because this environment has one CPU core. Benchmark run B0 [CL proposal], the first job before any production run: one 6-hour job with the

154-body setup, logging body count, collisions and wall time. The body count falls as bodies merge, and the cost per simulated year falls roughly with the square of the body count (286 bodies cost about 3.2 times as much as 154 in smoke test 2, close to the squared ratio of 3.4). B0 measures how fast that happens on the actual runner, so the total cost is projected from data instead of guessed.

Proposed settings [CL]:

- Starting disk: 154 bodies (14 embryos and 140 planetesimals between 0.3 and 2.0 AU), the Chambers (2013) setup used in [104]. Open link to D7: that setup places Jupiter and Saturn on their current orbits, which is only one of the giant-planet histories in the D7 selector. Each D7 scenario needs its own starting disk from its own mechanism source.
- Minimum fragment mass: half a planetesimal's mass, as in [104]. Smaller values capture more mantle-stripping collisions but add bodies and cost; [104] found that halving it did not change their results.
- Stopping rule: run until no collisions occur over a set window and the body count is stable, capped at 300 million years, the published run length in [104]. CBH sets the window length. A rule based on collisions, rather than a fixed time, stops finished runs early and saves compute.
- Repeat runs: 20 per scenario, matching the 20 concurrent jobs, against 25 to 43 in [104]. Results are reported as a spread. Only D11 scenario A can run until scenario B has its source.
- Expansion factor: see D12 below.

Caution found in [102]: inflating body radii speeds up collisions, but Childs and Steffen found that as the expansion factor rises, so does the average water fraction of the final planets. Since this model is about water, inflation biases the very output being tested.

Decision D12 for CBH: radius inflation (expansion factor) in the formation runs. A, no inflation (factor 1): physically clean and slowest; whether full-length runs fit free compute is what B0 will show. B, factor 3, the largest value reported in [104] as not changing outcomes significantly: faster, but water results carry the bias above. C, factor 1 for all production runs, plus a small factor-3 comparison set that measures the bias directly. [CL recommendation: A for now, revisited after B0 measures the real cost; move to C only if B0 shows factor 1 cannot finish.] D12 discussion notes, 4 October 2026 [CL]: three effects of radius inflation bear on this model. First, water: [102] found the final planets' average water fraction rises with the expansion factor. Second, collision outcomes: a check of the module code (fragmentation.c) shows it computes the mutual escape speed, the gravitational speed-up before impact, and the impact geometry from each body's stored radius, so with inflation the outcome category (merge, hit-and-run, erosion) is decided using the inflated size. That bears directly on mantle stripping and test F3. Whether [102] corrects for this was not checked. Third, timing: inflation shortens formation times by design, so formation ages cannot be compared with measured ones, such as Mars reaching near its final size within a few million years [14], without a correction. The case for inflation is compute alone.

Encounter check [CL], 4 October 2026: 2,100 two-body encounters (target 0.1 Earth mass; projectile 0.1, 0.3 and 1 times the target; approach speeds from 0.05 to 2 times the true mutual escape speed; aim points spread over the inflated target area) were run through the module's own collision code in REBOUND 4.4.10, once at true size and once with radii inflated 3 times, from identical starting conditions. Of the 409 encounters that collided at both sizes, impact speed measured against escape speed rose from a median of 1.27 to 1.65, and 270 (66 percent) changed outcome category. The largest shifts: elastic bounce to partial accretion (99), graze-and-merge to effectively merged (48), elastic bounce to effectively merged (44),

graze-and-merge to partial accretion (40), and hit-and-run to partial accretion (19). For the same encounter, inflation moves outcomes from glancing toward head-on and accreting. Inflation also turns near misses into collisions, and about half of all inflated collisions were scored as elastic bounces, where the bodies rebound with no mass exchanged. The share of collisions scored as hit-and-run moved in opposite directions by mass ratio (equal masses: 13 to 16 percent; ratio 0.3: 1.4 to 0.7 percent). Conclusion [CL]: inflation changes most individual collision outcomes, so its effect on mantle stripping cannot be removed with a simple correction. Source and reproduction steps: `encounter_check_v1.txt` in CBH's Google Drive.

Possible units issue found [CL, unconfirmed]: in the module's hit-and-run branch, the interacting fraction of the target (beta, labelled Chambers Eq. 11 in the code) is computed as interacting volume divided by target mass, with no density. In the check it ranged from about 10^{-11} to 3×10^{-6} , where a fraction of roughly 0.01 to 0.6 is expected; including the target's density gave values in that range. Rerunning with density included changed only the equal-mass hit-and-run counts (17 to 13 at true size, 111 to 76 inflated). Not yet checked against Chambers (2013) itself; worth raising with the module's author before production runs.

Units issue update [CL], 4 October 2026: the module's own paper [102] defines beta as the fraction of the target that would intersect the projectile in a head-on collision, a pure number.

The code's beta has units of volume over mass, so as written it cannot be that fraction; including density, as in the check, makes it one. This confirms the issue against [102]'s definition.

Chambers (2013) Eq. 11 itself is still unread, since it is not openly available.

Expansion-factor evidence from [102] [CL]: the authors used a factor of 3 as their smallest value for compute reasons. A factor-3 run took two weeks for 100 million years, while a factor-1 run took a week and a half for 5 million years and would need hundreds of millions of years to finish. So [102] never ran factor 1 to completion, and its data cannot show whether factor 3 differs from factor 1. They report that higher factors raise the merger rate, give more planets on wider orbits, and reduce radial mixing. They cite earlier work finding factors up to 10 have little effect, but under the condition that the bodies' random speeds are not dominated by gravitational scattering, a condition the late giant-impact stage may not meet.

Departing bodies [point raised by CBH; code check CL]: bodies leaving a collision are slowed by their mutual gravity. The module already includes this: fragments leave at slightly above escape speed, minus the speed lost climbing from contact to four combined radii. That slowing is reversible, stored as position energy rather than lost. The energy that is lost goes into heating and deforming the bodies at contact, which is what turns a would-be hit-and-run into a graze-and-merge in the module's rule.

Hosting decided [CBH, 4 October 2026]: both. GitHub Actions runs benchmark B0 and the production batches; Colab is the work area for inspecting results, running the core and mantle tracker [103], and making plots, with outputs saved to Drive. CBH has a paid Colab tier and Google AI Ultra. A single unofficial report says Ultra adds Colab hardware priority and background execution while compute units still cap use; not confirmed against Google's own pages, so not relied on.

Code files (the workflow, the problem file and the restart logic) will live in the GitHub repository, not in this document. This document will record their versions and hashes for the stage 5 freeze.

D12 decision and matched comparison P-D12

D12 decided [CBH, 4 October 2026]: radius inflation is settled by a preregistered matched comparison (P-D12) instead of being assumed or banned. Factor 3 is used in production only if

P-D12 shows it equivalent to true size under the committed rule below. Otherwise production runs at true size with a reduced scope, or at factor 3 with the measured bias reported beside every formation result; that choice comes back to CBH.

Preregistration draft [CL, pending CBH review; not yet certified]. It follows the commit-before-run discipline. The freeze block (file hashes) cannot be filled until the analysis script exists, and no P-D12 run may start before the filled preregistration passes the certification check and is committed to the repository.

- Classification: confirmatory. Question: up to a matched stage of growth, does factor 3 change the amount of mantle-stripping that collisions do, compared with true size?
- Setup: the 154-body Chambers (2013) disk as used in [104], Jupiter and Saturn on current orbits (D7 caveat applies), REBOUND 4.4.10 with TRACE, 6-day step, density 3 g/cm³, minimum fragment mass half a planetesimal mass, fragmentation module at commit d67ab2b unmodified. If the units fix is later adopted, P-D12 must be rerun as a new instrument.
- Pairs: 20 pairs, each pair sharing one random seed for its starting orbits, one run at factor 1 and one at factor 3. Unit of analysis: the run pair, so $n = 20$. Collisions within a run are not independent and are never pooled across runs for testing.
- Matched stage: each run is read at the moment its loose material (mass in bodies under 0.1 Earth mass, the yardstick of [102]) first falls to a fixed fraction of its starting value. The fraction is chosen from exploratory benchmark B0, whose seed is never reused in P-D12, and is written into the preregistration before certification.
- Primary statistic S: mass sent into fragments by stripping outcomes (hit-and-run, partial erosion, grazing partial erosion, super-catastrophic) up to the matched stage, divided by starting disk mass.
- Secondary statistic F: the fraction of a run's collisions that are stripping outcomes.
- Decision rule [CL, ~~margins pending CBH~~ margins approved by CBH, 4 October 2026]: factor 3 is ACCEPTED iff (1) the 90% bootstrap interval, over pairs, of the geometric-mean ratio $S(\text{factor } 3)/S(\text{factor } 1)$ lies entirely inside 0.8 to 1.25, AND (2) the 90% bootstrap interval of the paired difference in F lies entirely inside minus 0.05 to plus 0.05. Otherwise factor 3 is NOT ACCEPTED. An interval too wide to fit is a not-accepted result, not a pass: equivalence must be shown, not assumed.
- Same-condition control: 10 extra factor-1 runs with new seeds, each paired with one of the 20 factor-1 runs and scored by the same rule. If that factor-1 versus factor-1 comparison also fails the rule, the test cannot tell inflation from ordinary run-to-run scatter, and the verdict is INCONCLUSIVE rather than NOT ACCEPTED.
- Reproducibility ceiling: one factor-1 run repeated with the same seed must reproduce its collision report exactly; if not, the instrument is not trusted and P-D12 stops.
- Publish either way: the verdict, accepted or not, goes into this document and the repository README at equal prominence. A not-accepted result matters because the module's own authors could not run true size to completion, so in this toolchain inflation has not been tested against true size with fragmentation.
- Cheapest falsifier of an acceptance: before adopting factor 3, 5 extra pairs on the second Chambers (2013) disk used in [104] must also pass leg (1).
- Pre-freeze history: the encounter check of 4 October 2026 was exploratory and shaped the choice of S and F; B0 is exploratory; neither is reported as evidence for or against factor 3.
- Two ways this is still fooling us [CL]: first, matching by remaining loose material ignores that orbits heat up with real time, so factor-3 runs reach the stage with calmer orbits; an

acceptance covers only the stage tested, not the late giant impacts where Mercury's stripping may happen. Second, the margins 0.8 to 1.25 and plus or minus 0.05 were picked by Claude, not derived from what F3 can tolerate; a smaller bias might still flip F3.

Open items for CBH

- Review the process inventory (P1 to P26) and the coupling table (C1 to C19).
- ~~Next block: verify equation forms against sources, starting with escape (P18 to P20) and convection (P7, P8).~~
- ~~Next block: verify E6 dynamo criterion (P8) and E7 D/H enrichment (P26), then find sources for P9, P10, P13 and P21; run the planned checks for E4 and E5.~~
- ~~Next block (3): greenhouse warming (P12), weathering rate (P14), stagnant lid outgassing (P9 for Venus, Mars and Mercury), core formation (P4), accretion parameters (P1); run the planned checks for E4, E5 and E8.~~
- ~~Next block (4): accretion parameters (P1), core formation (P4), thick CO₂ greenhouse for Venus and early Mars, stagnant lid outgassing equations from FS25, the young Sun's EUV exponent checked in ref 18 directly, and the pending E4 and E5 checks.~~
- ~~Decision D10 for CBH: how the model computes accretion outcomes (options A, B, C in block 4). Then block 5: remaining coefficients (thick CO₂ fit, single-shell melt fraction), building block iron content for core formation.~~
- ~~Next: design the accretion runs (body count, duration, realizations, collision outcome model with a verified source), then block 5: thick CO₂ coefficients, single-shell melt fraction, building block iron content.~~
- ~~Next: CBH decides D11 (how building block iron is set), then accretion run design part 2 (run length, repeat runs, body count, minimum fragment mass, hosting under the compute limit from smoke test 2), then block 5 (thick CO₂ coefficients, single-shell melt fraction).~~
- ~~Next: accretion run design part 2 (run length, repeat runs, body count, minimum fragment mass, hosting under the compute limit from smoke test 2), including a verified mechanism source with a number for scenario B's iron gradient; then block 5 (thick CO₂ coefficients, single-shell melt fraction).~~
- ~~Next: CBH decides D12 (radius inflation) and approves the hosting plan; then set up the public repository and run benchmark B0; then block 5 (thick CO₂ coefficients, single-shell melt fraction). The search for scenario B's iron gradient source runs alongside.~~
- ~~Next: CBH decides D12 (radius inflation, under discussion); then set up the public repository and run benchmark B0; then block 5 (thick CO₂ coefficients, single-shell melt fraction). The search for scenario B's iron gradient source runs alongside.~~
- ~~Next: CBH reviews the P-D12 draft (especially the equivalence margins); then set up the public repository, run exploratory benchmark B0, write and freeze the P-D12 analysis script, certify the preregistration, and run P-D12; then block 5 (thick CO₂ coefficients, single-shell melt fraction). Alongside: the search for scenario B's iron gradient source, and whether to raise the units issue with the module's author.~~
- Next: set up the public repository (CBH creates an empty public repo under NavigatorsLog; Claude prepares the files), run exploratory benchmark B0, write and freeze the P-D12 analysis script, certify the preregistration, and run P-D12; then block 5 (thick-CO₂ coefficients, single-shell melt fraction). Alongside: the search for scenario B's iron-gradient source, and whether to raise the units issue with the module's author (CBH: not yet).

Revision log

v0.1, 3 October 2026: started; process inventory and coupling table drafted [CL].

v0.1, 3 October 2026, second entry: verified equation block 1 (E1 to E5) added with sources for each form and numerical checks; block 2 queued [CL].

v0.1, 3 October 2026, third entry: verified equation block 2 (E6 to E10) added with sources and checks; open gaps narrowed to stagnant-lid outgassing and the block 3 list [CL].

v0.1, 3 October 2026, fourth entry: verified equation block 3 (E11 to E14), the E8 reproduction check, and proposed couplings C20 to C22 [CL]. Block 4 queued.

v0.1, 3 October 2026, fifth entry: block 4 added (E14 upgraded to partly verified, E15 and E16, accretion and core formation status); E8 caveat closed; decision D10 on accretion raised for CBH [CL].

v0.1, 3 October 2026, sixth entry: CBH decided D10 = A (own N-body accretion runs); feasibility smoke test recorded with compute estimate and open design choices [CL].

v0.1, 4 October 2026, seventh entry: accretion run design part 1 added (collision-outcome model with sources, refs 99 to 105 in the Specification; smoke test 2 with timing; the F3 finding from $\diamond 104$; decision D11) [CL, pending CBH]. The previous Next item in Open items is struck through, not deleted.

v0.1, 4 October 2026, eighth entry: CBH decided D11 = C (building-block iron runs as two named scenarios, F3 reported for each). The previous Next item is struck through.

v0.1, 4 October 2026, ninth entry: accretion run design part 2 added (hosting plan on GitHub Actions, benchmark B0, proposed settings, the expansion-factor caution from [102], decision D12) [CL, pending CBH]. The previous Next item is struck through.

v0.1, 4 October 2026, tenth entry: hosting decided by CBH (GitHub Actions for runs, Colab for analysis); D12 discussion notes added, including a code check of how the module uses body radius [CL]. D12 stays open. The previous Next item is struck through.

v0.1, 4 October 2026, eleventh entry: encounter check for D12 added (2,100 paired encounters through the module's collision code), a possible units issue in the module's hit-and-run branch [CL, unconfirmed], and the departing-bodies note raised by CBH. Harness saved to Drive as `encounter_check_v1.txt`.

v0.1, 4 October 2026, twelfth entry: the units issue checked against [102]'s definition of beta (confirmed there; Chambers 2013 still unread), and expansion-factor evidence from [102] added to the D12 notes [CL].

v0.1, 4 October 2026, thirteenth entry: CBH decided D12 (matched comparison P-D12 settles radius inflation); preregistration draft for P-D12 added [CL, pending CBH review]. The previous Next item is struck through.

v0.1, 4 October 2026, fourteenth entry: CBH approved the P-D12 equivalence margins as drafted; CBH chose not to contact the module's author yet. The old margin label and the previous Next item are struck through.

v0.1, 4 October 2026, fifteenth entry: repository files drafted [CL] and saved to CBH's Drive folder "nl-formation-runs (repo draft, 4 Oct 2026)": `README.md`, `.gitignore`, `scripts/setup.sh` (fetches REBOUND 4.4.10 and the module at d67ab2b, builds, records SHA-256 hashes), `src/formation_run.c` (154-body Chambers 2013 bimodal disk, TRACE, wall-clock stop with save and resume), `.github/workflows/b0.yml` (B0 as two GitHub Actions jobs, factors 1 and 3, seed 900001), `analysis/b0_summary.py` (exploratory). Tested here: builds without errors; a wall-clock stop saves cleanly; a resumed run reproduces the same collision sequence as an uninterrupted

run with the same seed. Two inputs are inferred, not read from a source, and are flagged in the code for checking against Chambers (2013): body masses of 0.093 and 0.0093 Earth masses (from the published total of 2.604), and a surface density starting from zero at 0.3 AU (this profile reproduces the published total to within about 3 percent). Open for CBH: repository license (GPL-3.0 suggested, because the GPL-3.0 module is compiled in). The P-D12 analysis script is not yet written.